

DATABASE MANAGEMENT SYSTEM

Laboratory Equipment Booking & Maintenance System

Project 29 · Review 1

Presented By

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Agenda

01	02	03
Problem Identification	Scope	Objectives
Understanding the core challenges in lab equipment management	Defining the boundaries and coverage of the system	Goals the system aims to achieve
04	05	06
System Users	Functional Requirements	Conceptual Design / ER Diagram
Roles and responsibilities of each user type	Core features and capabilities required	Entity-relationship model of the system
07		
Initial Relational Schema		
Proposed tables with primary and foreign keys		



INTRODUCTION

System Overview

The Laboratory Equipment Booking and Maintenance System is a structured relational database designed to digitize and streamline the complete lifecycle of laboratory resource management in an academic or research institution.

Laboratories & Equipment

Tracks all labs, equipment categories, individual equipment items and their current status

Users & Bookings

Manages user profiles, time slot definitions, booking requests and approvals

Usage & Maintenance

Records usage logs, breakdown reports, maintenance jobs, calibration schedules and consumables

Problem Statement

Without a centralized system, laboratories face recurring operational challenges that hinder productivity, safety and equipment longevity.



Booking Conflicts

Overlapping reservations occur due to lack of real-time visibility into equipment availability, causing disruption to research schedules



Availability Tracking

No unified mechanism exists to check whether specific equipment is operational, under maintenance or already reserved



Weak Usage & Maintenance History

Usage records and maintenance histories are fragmented or paper-based, making it difficult to audit or plan preventive maintenance



Missed Calibrations & Poor Consumable Tracking

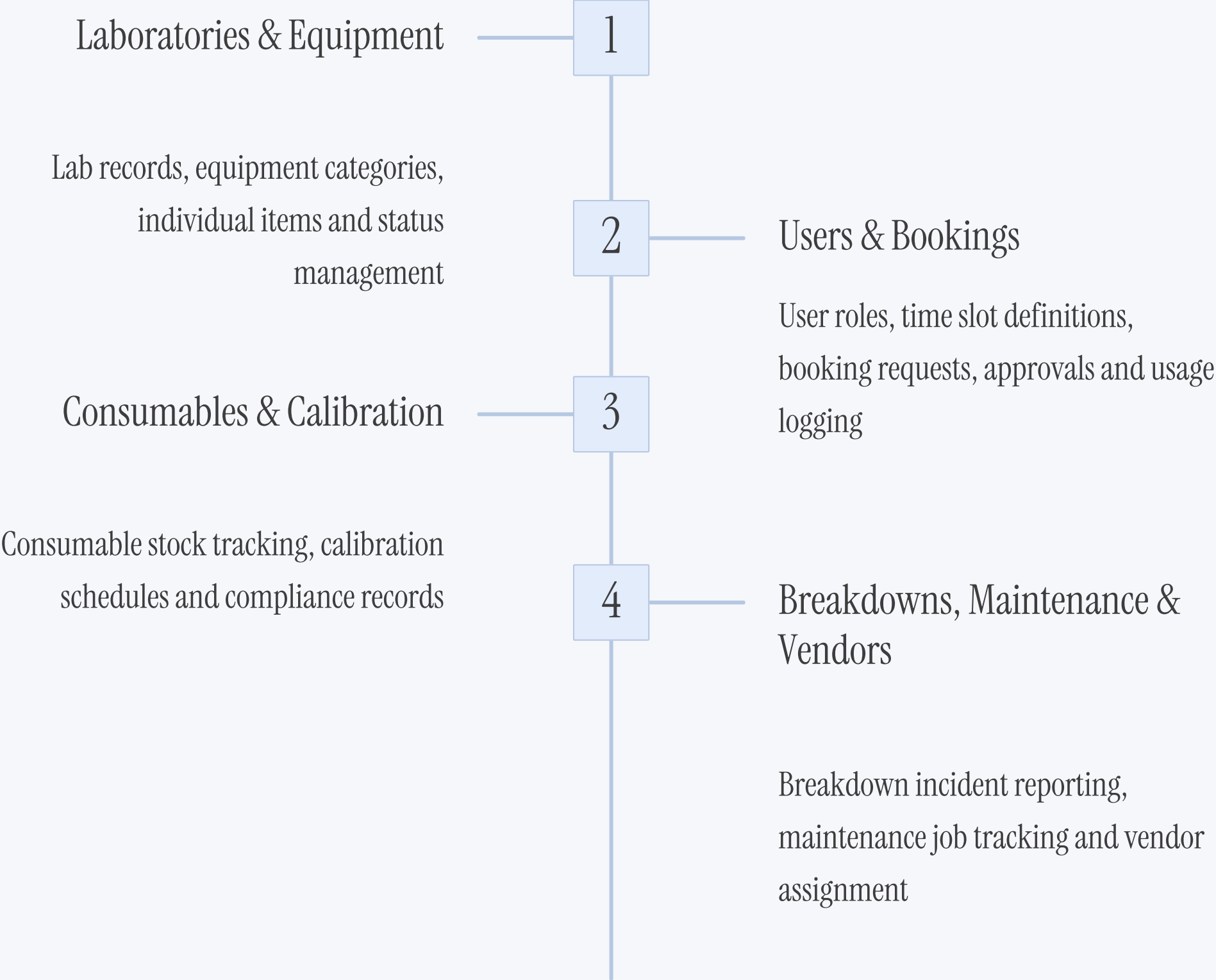
Calibration due dates are missed without automated reminders; consumable stock levels go unmonitored until shortages occur

SCOPE

System Scope

The system covers all key entities and processes needed to manage a multi-laboratory environment end to end.

Out-of-scope items include financial billing, procurement workflows and HR management.



System Objectives

01	02	03
Centralize Laboratory Data	Enable Equipment Search & Booking	Prevent Conflicting Approvals
Consolidate all labs, equipment, users and transactions into a single relational database for consistent and reliable access	Allow users to discover available equipment by category or lab and submit time-slot-based booking requests	Enforce constraints so no two approved bookings can overlap for the same equipment in the same time slot
04	05	
Restrict Bookings to Active Equipment	Track Usage, Calibration, Breakdowns & Consumables	
Only equipment with an "Active" status may be reserved; items under repair or decommissioned are automatically excluded	Maintain complete operational history and generate reports for audits, preventive maintenance planning and consumable replenishment	

User Roles & Responsibilities

Student

- Search for available equipment
- Submit booking requests for approved time slots
- Log usage after completing a session
- Report breakdowns encountered during use

Faculty

- Book equipment for research and teaching
- Approve or reject student booking requests
- View usage history for their assigned labs
- Initiate calibration requests

Technician

- Update equipment status after inspections
- Log and resolve breakdown incidents
- Record maintenance job details and outcomes
- Manage calibration schedules and consumable stock

Administrator

- Manage all users, labs and equipment records
- Configure time slots and booking policies
- Assign vendors for maintenance jobs
- Generate system-wide reports and audit logs

Functional Requirements

1 Equipment Management

Add, update and deactivate equipment records; maintain categories and lab assignments

2 Search & Filter

Query equipment by name, category, lab or availability status in real time

3 Booking & Approval Workflow

Submit requests, enforce no-overlap constraint on approved bookings and notify users of status changes

4 Usage Logging

Record start/end times, purpose and the user responsible for each equipment session

1 Breakdown Reporting

Log breakdown incidents with severity, description and reported date; link to affected equipment

2 Maintenance Management

Create maintenance jobs from breakdowns, assign vendors and track job status and completion dates

3 Calibration Tracking

Schedule calibrations, record results and flag overdue calibrations automatically

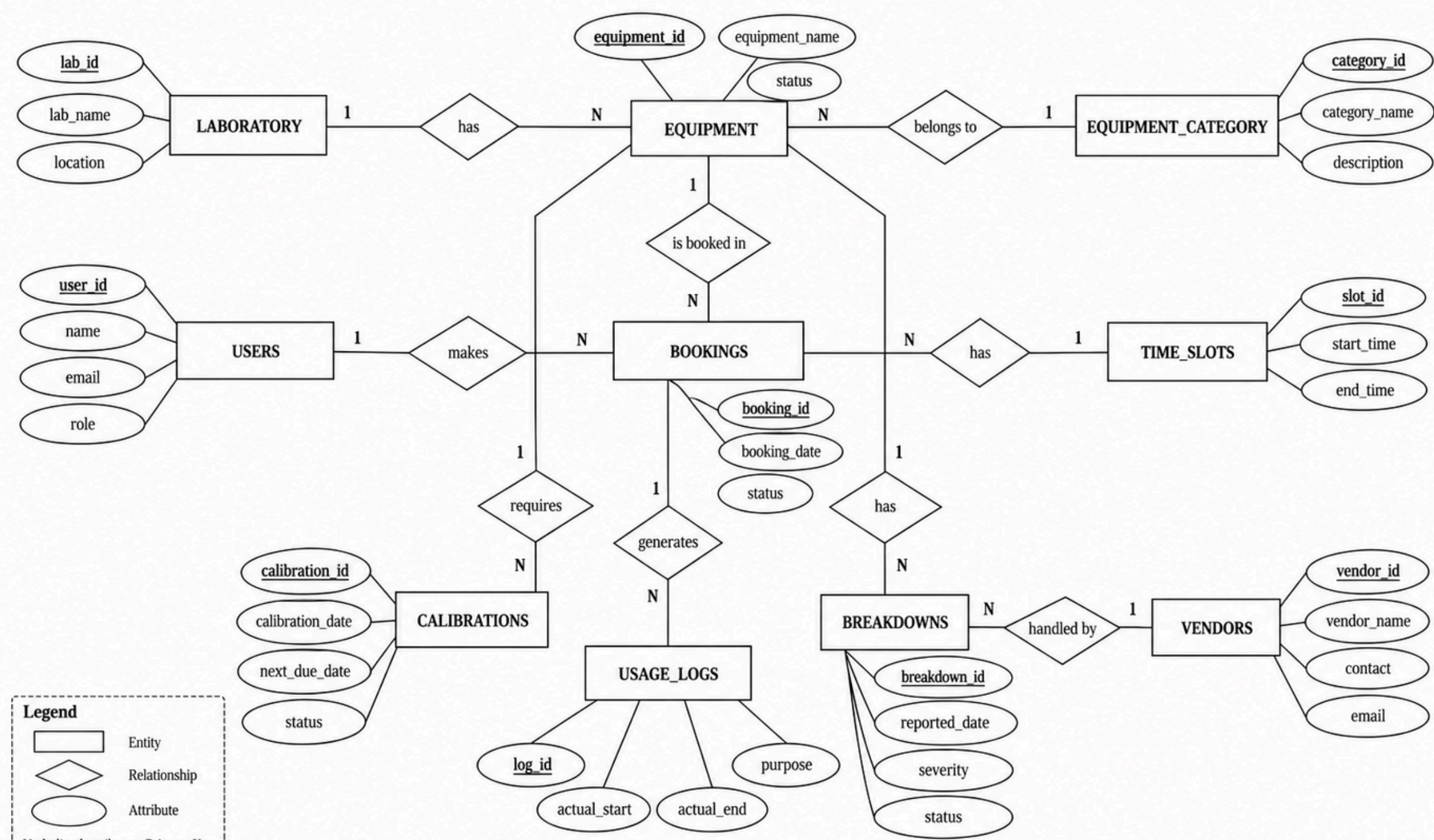
4 Consumable Management & Reports

Monitor stock quantities, record usage per session and generate operational and audit reports

ER Diagram – Entity Overview

Central entity Equipment connects to all operational records – Bookings, Usage Logs, Calibrations and Breakdowns – while Labs, Categories and Users provide organizational structure.

Laboratory Equipment Booking and Maintenance System
ER Diagram (Chen’s Notation)



Proposed Table Structures

Table	Primary Key	Key Foreign Keys
Laboratories	lab_id (PK)	—
Equipment_Categories	category_id (PK)	—
Equipment	equipment_id (PK)	lab_id (FK), category_id (FK)
Users	user_id (PK)	—
Time_Slots	slot_id (PK)	—
Bookings	booking_id (PK)	equipment_id (FK), user_id (FK), slot_id (FK)
Usage_Logs	log_id (PK)	booking_id (FK), equipment_id (FK), user_id (FK)
Consumables	consumable_id (PK)	lab_id (FK)
Calibrations	calibration_id (PK)	equipment_id (FK)
Breakdowns	breakdown_id (PK)	equipment_id (FK), reported_by (FK → Users)
Vendors	vendor_id (PK)	—
Maintenance_Jobs	job_id (PK)	breakdown_id (FK), vendor_id (FK), equipment_id (FK)



Thank You